# Reversed achievement score in primary school for the youngest child

Current 2022/2023 enrollment defines the primary school sample. One youngest child within this level is selected per household before subject-specific outcome exclusions. 6 minus the recorded response for responses 1 through 5. Higher scores indicate better achievement. Codes 6 and 99 and blanks are excluded separately for each subject. Father estimates are primary and mother estimates are secondary.

## Father RDD primary analysis

*Specification 1 Monthly child-age fixed effects  No donut hole*

| Outcome | Mean | 8 years | 7 years | 6 years | 5 years | 4 years |
|---|---|---|---|---|---|---|
| Turkish | 3.928 | 0.008 [0.066]<br>N=1,919 | 0.041 [0.068]<br>N=1,775 | 0.077 [0.073]<br>N=1,600 | 0.099 [0.079]<br>N=1,393 | 0.053 [0.088]<br>N=1,167 |
| Social studies and life sciences | 3.940 | 0.073 [0.058]<br>N=1,781 | **0.102* [0.061]**<br>**N=1,647** | **0.122* [0.066]**<br>**N=1,484** | **0.134* [0.073]**<br>**N=1,297** | 0.081 [0.084]<br>N=1,087 |
| Mathematics | 3.872 | 0.069 [0.069]<br>N=1,884 | **0.125* [0.072]**<br>**N=1,744** | **0.140* [0.076]**<br>**N=1,571** | **0.170** [0.083]**<br>**N=1,369** | 0.130 [0.094]<br>N=1,152 |
| Science | 3.872 | 0.018 [0.091]<br>N=1,286 | 0.055 [0.095]<br>N=1,182 | 0.090 [0.102]<br>N=1,056 | 0.099 [0.113]<br>N=919 | 0.036 [0.131]<br>N=768 |
| Foreign language | 3.688 | 0.083 [0.095]<br>N=1,424 | 0.151 [0.099]<br>N=1,312 | 0.144 [0.110]<br>N=1,184 | 0.144 [0.121]<br>N=1,034 | 0.144 [0.137]<br>N=866 |
| Culture arts and sports | 3.998 | 0.038 [0.062]<br>N=1,573 | 0.051 [0.066]<br>N=1,451 | 0.018 [0.070]<br>N=1,311 | -0.005 [0.080]<br>N=1,140 | -0.063 [0.087]<br>N=959 |
| Vocational and technological | 3.844 | 0.048 [0.135]<br>N=584 | 0.069 [0.142]<br>N=530 | 0.101 [0.155]<br>N=484 | 0.119 [0.159]<br>N=422 | 0.025 [0.176]<br>N=348 |

*Specification 1 Monthly child-age fixed effects  12 month donut hole*

| Outcome | 8 years | 7 years | 6 years | 5 years | 4 years |
|---|---|---|---|---|---|
| Turkish | -0.089 [0.083]<br>N=1,610 | -0.045 [0.091]<br>N=1,466 | 0.014 [0.102]<br>N=1,291 | 0.032 [0.112]<br>N=1,084 | -0.142 [0.127]<br>N=858 |
| Social studies and life sciences | 0.031 [0.079]<br>N=1,491 | 0.073 [0.086]<br>N=1,357 | 0.107 [0.098]<br>N=1,194 | 0.125 [0.106]<br>N=1,007 | -0.035 [0.117]<br>N=797 |
| Mathematics | 0.051 [0.092]<br>N=1,579 | 0.146 [0.100]<br>N=1,439 | 0.177 [0.111]<br>N=1,266 | **0.241* [0.126]**<br>**N=1,064** | 0.163 [0.161]<br>N=847 |
| Science | -0.104 [0.109]<br>N=1,088 | -0.055 [0.119]<br>N=984 | -0.011 [0.128]<br>N=858 | -0.047 [0.153]<br>N=721 | -0.239 [0.189]<br>N=570 |
| Foreign language | -0.013 [0.120]<br>N=1,204 | 0.104 [0.131]<br>N=1,092 | 0.059 [0.147]<br>N=964 | 0.056 [0.166]<br>N=814 | -0.008 [0.192]<br>N=646 |
| Culture arts and sports | 0.070 [0.095]<br>N=1,325 | 0.092 [0.106]<br>N=1,203 | 0.043 [0.118]<br>N=1,063 | 0.034 [0.150]<br>N=892 | -0.096 [0.190]<br>N=711 |
| Vocational and technological | 0.115 [0.185]<br>N=503 | 0.181 [0.200]<br>N=449 | 0.244 [0.233]<br>N=403 | 0.295 [0.252]<br>N=341 | 0.189 [0.337]<br>N=267 |

*Specification 2 Monthly fixed effects and female  No donut hole*

| Outcome | Mean | 8 years | 7 years | 6 years | 5 years | 4 years |
|---|---|---|---|---|---|---|
| Turkish | 3.928 | 0.007 [0.065]<br>N=1,919 | 0.041 [0.067]<br>N=1,775 | 0.074 [0.072]<br>N=1,600 | 0.099 [0.079]<br>N=1,393 | 0.058 [0.087]<br>N=1,167 |
| Social studies and life sciences | 3.940 | 0.072 [0.057]<br>N=1,781 | **0.101* [0.060]<br>N=1,647** | **0.119* [0.066]<br>N=1,484** | **0.134* [0.073]<br>N=1,297** | 0.085 [0.084]<br>N=1,087 |
| Mathematics | 3.872 | 0.068 [0.070]<br>N=1,884 | **0.125* [0.072]<br>N=1,744** | **0.141* [0.077]<br>N=1,571** | **0.169** [0.083]<br>N=1,369** | 0.128 [0.094]<br>N=1,152 |
| Science | 3.872 | 0.014 [0.091]<br>N=1,286 | 0.051 [0.094]<br>N=1,182 | 0.082 [0.100]<br>N=1,056 | 0.095 [0.112]<br>N=919 | 0.038 [0.129]<br>N=768 |
| Foreign language | 3.688 | 0.081 [0.093]<br>N=1,424 | 0.149 [0.098]<br>N=1,312 | 0.140 [0.108]<br>N=1,184 | 0.142 [0.119]<br>N=1,034 | 0.151 [0.134]<br>N=866 |
| Culture arts and sports | 3.998 | 0.038 [0.062]<br>N=1,573 | 0.050 [0.066]<br>N=1,451 | 0.017 [0.070]<br>N=1,311 | -0.004 [0.080]<br>N=1,140 | -0.060 [0.087]<br>N=959 |
| Vocational and technological | 3.844 | 0.042 [0.135]<br>N=584 | 0.064 [0.143]<br>N=530 | 0.085 [0.156]<br>N=484 | 0.103 [0.160]<br>N=422 | 0.020 [0.176]<br>N=348 |

*Specification 2 Monthly fixed effects and female  12 month donut hole*

| Outcome | 8 years | 7 years | 6 years | 5 years | 4 years |
|---|---|---|---|---|---|
| Turkish | -0.096 [0.084]<br>N=1,610 | -0.052 [0.092]<br>N=1,466 | -0.000 [0.103]<br>N=1,291 | 0.023 [0.114]<br>N=1,084 | -0.140 [0.128]<br>N=858 |
| Social studies and life sciences | 0.026 [0.079]<br>N=1,491 | 0.067 [0.086]<br>N=1,357 | 0.095 [0.099]<br>N=1,194 | 0.118 [0.107]<br>N=1,007 | -0.029 [0.118]<br>N=797 |
| Mathematics | 0.053 [0.092]<br>N=1,579 | 0.147 [0.100]<br>N=1,439 | 0.179 [0.112]<br>N=1,266 | **0.245* [0.126]<br>N=1,064** | 0.163 [0.161]<br>N=847 |
| Science | -0.108 [0.111]<br>N=1,088 | -0.059 [0.121]<br>N=984 | -0.024 [0.130]<br>N=858 | -0.050 [0.154]<br>N=721 | -0.228 [0.186]<br>N=570 |
| Foreign language | -0.019 [0.118]<br>N=1,204 | 0.100 [0.129]<br>N=1,092 | 0.049 [0.145]<br>N=964 | 0.049 [0.164]<br>N=814 | 0.008 [0.189]<br>N=646 |
| Culture arts and sports | 0.066 [0.095]<br>N=1,325 | 0.088 [0.106]<br>N=1,203 | 0.038 [0.118]<br>N=1,063 | 0.028 [0.150]<br>N=892 | -0.093 [0.189]<br>N=711 |
| Vocational and technological | 0.100 [0.189]<br>N=503 | 0.168 [0.205]<br>N=449 | 0.209 [0.241]<br>N=403 | 0.275 [0.258]<br>N=341 | 0.187 [0.339]<br>N=267 |

*Specification 3 Monthly fixed effects female and household income groups No donut hole*

| Outcome | Mean | 8 years | 7 years | 6 years | 5 years | 4 years |
|---|---|---|---|---|---|---|
| Turkish | 3.928 | 0.001 [0.064]<br>N=1,919 | 0.031 [0.066]<br>N=1,775 | 0.061 [0.070]<br>N=1,600 | 0.079 [0.076]<br>N=1,393 | 0.031 [0.083]<br>N=1,167 |
| Social studies and life sciences | 3.940 | 0.065 [0.057]<br>N=1,781 | 0.088 [0.060]<br>N=1,647 | 0.101 [0.066]<br>N=1,484 | 0.108 [0.072]<br>N=1,297 | 0.048 [0.083]<br>N=1,087 |
| Mathematics | 3.872 | 0.059 [0.068]<br>N=1,884 | 0.110 [0.071]<br>N=1,744 | 0.119 [0.075]<br>N=1,571 | **0.139* [0.080]<br>N=1,369** | 0.095 [0.090]<br>N=1,152 |
| Science | 3.872 | -0.014 [0.088]<br>N=1,286 | 0.015 [0.092]<br>N=1,182 | 0.032 [0.099]<br>N=1,056 | 0.037 [0.110]<br>N=919 | -0.018 [0.126]<br>N=768 |
| Foreign language | 3.688 | 0.078 [0.093]<br>N=1,424 | 0.136 [0.098]<br>N=1,312 | 0.126 [0.108]<br>N=1,184 | 0.123 [0.120]<br>N=1,034 | 0.132 [0.136]<br>N=866 |
| Culture arts and sports | 3.998 | 0.032 [0.060]<br>N=1,573 | 0.040 [0.064]<br>N=1,451 | 0.002 [0.067]<br>N=1,311 | -0.029 [0.076]<br>N=1,140 | -0.089 [0.084]<br>N=959 |
| Vocational and technological | 3.844 | 0.013 [0.137]<br>N=584 | 0.007 [0.147]<br>N=530 | 0.000 [0.162]<br>N=484 | 0.022 [0.166]<br>N=422 | -0.057 [0.183]<br>N=348 |

*Specification 3 Monthly fixed effects female and household income groups 12 month donut hole*

| Outcome | 8 years | 7 years | 6 years | 5 years | 4 years |
|---|---|---|---|---|---|
| Turkish | -0.111 [0.084]<br>N=1,610 | -0.076 [0.092]<br>N=1,466 | -0.028 [0.103]<br>N=1,291 | -0.034 [0.114]<br>N=1,084 | **-0.242* [0.126]<br>N=858** |
| Social studies and life sciences | 0.019 [0.080]<br>N=1,491 | 0.049 [0.087]<br>N=1,357 | 0.066 [0.101]<br>N=1,194 | 0.060 [0.109]<br>N=1,007 | -0.141 [0.124]<br>N=797 |
| Mathematics | 0.031 [0.092]<br>N=1,579 | 0.113 [0.103]<br>N=1,439 | 0.132 [0.115]<br>N=1,266 | 0.165 [0.126]<br>N=1,064 | 0.047 [0.165]<br>N=847 |
| Science | -0.130 [0.108]<br>N=1,088 | -0.092 [0.119]<br>N=984 | -0.083 [0.131]<br>N=858 | -0.137 [0.155]<br>N=721 | **-0.320* [0.187]<br>N=570** |
| Foreign language | -0.027 [0.119]<br>N=1,204 | 0.074 [0.131]<br>N=1,092 | 0.020 [0.150]<br>N=964 | -0.008 [0.166]<br>N=814 | -0.068 [0.203]<br>N=646 |
| Culture arts and sports | 0.053 [0.095]<br>N=1,325 | 0.063 [0.106]<br>N=1,203 | -0.002 [0.117]<br>N=1,063 | -0.058 [0.147]<br>N=892 | -0.234 [0.185]<br>N=711 |
| Vocational and technological | 0.124 [0.195]<br>N=503 | 0.155 [0.215]<br>N=449 | 0.170 [0.255]<br>N=403 | 0.275 [0.280]<br>N=341 | 0.224 [0.357]<br>N=267 |

*Specification 4 Monthly fixed effects female and per-person income groups  No donut hole*

| Outcome | Mean | 8 years | 7 years | 6 years | 5 years | 4 years |
|---|---|---|---|---|---|---|
| Turkish | 3.928 | -0.002 [0.063]<br>N=1,919 | 0.023 [0.065]<br>N=1,775 | 0.046 [0.068]<br>N=1,600 | 0.062 [0.073]<br>N=1,393 | 0.009 [0.080]<br>N=1,167 |
| Social studies and life sciences | 3.940 | 0.059 [0.056]<br>N=1,781 | 0.078 [0.059]<br>N=1,647 | 0.083 [0.064]<br>N=1,484 | 0.087 [0.069]<br>N=1,297 | 0.023 [0.080]<br>N=1,087 |
| Mathematics | 3.872 | 0.055 [0.067]<br>N=1,884 | 0.102 [0.071]<br>N=1,744 | 0.106 [0.075]<br>N=1,571 | 0.129 [0.079]<br>N=1,369 | 0.077 [0.089]<br>N=1,152 |
| Science | 3.872 | -0.014 [0.087]<br>N=1,286 | 0.011 [0.091]<br>N=1,182 | 0.022 [0.097]<br>N=1,056 | 0.028 [0.107]<br>N=919 | -0.043 [0.122]<br>N=768 |
| Foreign language | 3.688 | 0.073 [0.090]<br>N=1,424 | 0.131 [0.094]<br>N=1,312 | 0.114 [0.104]<br>N=1,184 | 0.106 [0.114]<br>N=1,034 | 0.109 [0.128]<br>N=866 |
| Culture arts and sports | 3.998 | 0.024 [0.060]<br>N=1,573 | 0.028 [0.064]<br>N=1,451 | -0.017 [0.067]<br>N=1,311 | -0.042 [0.075]<br>N=1,140 | -0.110 [0.082]<br>N=959 |
| Vocational and technological | 3.844 | 0.049 [0.131]<br>N=584 | 0.032 [0.140]<br>N=530 | 0.031 [0.152]<br>N=484 | 0.052 [0.156]<br>N=422 | -0.041 [0.173]<br>N=348 |

*Specification 4 Monthly fixed effects female and per-person income groups  12 month donut hole*

| Outcome | 8 years | 7 years | 6 years | 5 years | 4 years |
|---|---|---|---|---|---|
| Turkish | -0.116 [0.081]<br>N=1,610 | -0.090 [0.088]<br>N=1,466 | -0.054 [0.100]<br>N=1,291 | -0.070 [0.107]<br>N=1,084 | **-0.291** [0.116]<br>N=858** |
| Social studies and life sciences | 0.008 [0.077]<br>N=1,491 | 0.028 [0.085]<br>N=1,357 | 0.032 [0.099]<br>N=1,194 | 0.016 [0.103]<br>N=1,007 | **-0.198* [0.114]<br>N=797** |
| Mathematics | 0.029 [0.092]<br>N=1,579 | 0.100 [0.102]<br>N=1,439 | 0.110 [0.115]<br>N=1,266 | 0.150 [0.126]<br>N=1,064 | 0.027 [0.165]<br>N=847 |
| Science | -0.133 [0.108]<br>N=1,088 | -0.107 [0.120]<br>N=984 | -0.107 [0.132]<br>N=858 | -0.156 [0.151]<br>N=721 | **-0.359* [0.180]<br>N=570** |
| Foreign language | -0.029 [0.115]<br>N=1,204 | 0.069 [0.127]<br>N=1,092 | 0.001 [0.142]<br>N=964 | -0.034 [0.158]<br>N=814 | -0.098 [0.190]<br>N=646 |
| Culture arts and sports | 0.044 [0.097]<br>N=1,325 | 0.045 [0.108]<br>N=1,203 | -0.037 [0.119]<br>N=1,063 | -0.074 [0.152]<br>N=892 | -0.268 [0.190]<br>N=711 |
| Vocational and technological | 0.170 [0.190]<br>N=503 | 0.187 [0.210]<br>N=449 | 0.209 [0.244]<br>N=403 | 0.306 [0.268]<br>N=341 | 0.261 [0.346]<br>N=267 |

*Specification 5 Monthly fixed effects female and number of aid types (0-6) No donut hole*

| Outcome | Mean | 8 years | 7 years | 6 years | 5 years | 4 years |
|---|---|---|---|---|---|---|
| Turkish | 3.916 | -0.027 [0.067] N=1,806 | -0.000 [0.070] N=1,670 | 0.032 [0.075] N=1,503 | 0.051 [0.082] N=1,310 | -0.003 [0.092] N=1,096 |
| Social studies and life sciences | 3.929 | 0.063 [0.058] N=1,677 | 0.084 [0.061] N=1,549 | 0.101 [0.066] N=1,393 | 0.112 [0.074] N=1,219 | 0.054 [0.085] N=1,021 |
| Mathematics | 3.862 | 0.044 [0.074] N=1,774 | 0.095 [0.078] N=1,640 | 0.103 [0.083] N=1,475 | 0.130 [0.092] N=1,287 | 0.082 [0.102] N=1,082 |
| Science | 3.859 | 0.009 [0.088] N=1,207 | 0.040 [0.093] N=1,107 | 0.073 [0.100] N=987 | 0.097 [0.110] N=860 | 0.048 [0.127] N=716 |
| Foreign language | 3.670 | 0.051 [0.101] N=1,343 | 0.129 [0.107] N=1,236 | 0.119 [0.117] N=1,113 | 0.135 [0.130] N=974 | 0.141 [0.147] N=812 |
| Culture arts and sports | 3.996 | 0.045 [0.063] N=1,508 | 0.058 [0.067] N=1,390 | 0.020 [0.069] N=1,256 | 0.008 [0.078] N=1,095 | -0.038 [0.087] N=919 |
| Vocational and technological | 3.837 | 0.075 [0.135] N=563 | 0.095 [0.144] N=512 | 0.125 [0.160] N=466 | 0.126 [0.170] N=411 | 0.035 [0.185] N=338 |

*Specification 5 Monthly fixed effects female and number of aid types (0-6) 12 month donut hole*

| Outcome | 8 years | 7 years | 6 years | 5 years | 4 years |
|---|---|---|---|---|---|
| Turkish | -0.093 [0.086] N=1,520 | -0.057 [0.093] N=1,384 | -0.004 [0.104] N=1,217 | 0.007 [0.117] N=1,024 | -0.151 [0.138] N=810 |
| Social studies and life sciences | 0.043 [0.080] N=1,408 | 0.073 [0.086] N=1,280 | 0.099 [0.098] N=1,124 | 0.116 [0.111] N=950 | -0.014 [0.122] N=752 |
| Mathematics | 0.053 [0.098] N=1,492 | 0.142 [0.106] N=1,358 | 0.163 [0.119] N=1,193 | **0.233* [0.139] N=1,005** | 0.161 [0.174] N=800 |
| Science | -0.108 [0.109] N=1,026 | -0.074 [0.120] N=926 | -0.043 [0.132] N=806 | -0.066 [0.156] N=679 | -0.196 [0.186] N=535 |
| Foreign language | -0.039 [0.125] N=1,143 | 0.082 [0.137] N=1,036 | 0.025 [0.154] N=913 | 0.035 [0.176] N=774 | -0.005 [0.201] N=612 |
| Culture arts and sports | 0.060 [0.099] N=1,274 | 0.079 [0.110] N=1,156 | 0.018 [0.121] N=1,022 | 0.009 [0.152] N=861 | -0.085 [0.197] N=685 |
| Vocational and technological | 0.183 [0.171] N=487 | 0.232 [0.186] N=436 | 0.291 [0.231] N=390 | 0.332 [0.267] N=335 | 0.197 [0.341] N=262 |

*Specification 6 Monthly fixed effects female and any aid (0/1) No donut hole*

| Outcome | Mean | 8 years | 7 years | 6 years | 5 years | 4 years |
|---|---|---|---|---|---|---|
| Turkish | 3.916 | -0.032 [0.068] N=1,806 | 0.002 [0.071] N=1,670 | 0.034 [0.076] N=1,503 | 0.052 [0.083] N=1,310 | -0.001 [0.093] N=1,096 |
| Social studies and life sciences | 3.929 | 0.058 [0.059] N=1,677 | 0.085 [0.061] N=1,549 | 0.102 [0.067] N=1,393 | 0.110 [0.074] N=1,219 | 0.049 [0.086] N=1,021 |
| Mathematics | 3.862 | 0.039 [0.075] N=1,774 | 0.097 [0.079] N=1,640 | 0.104 [0.084] N=1,475 | 0.130 [0.093] N=1,287 | 0.083 [0.103] N=1,082 |
| Science | 3.859 | -0.004 [0.090] N=1,207 | 0.033 [0.094] N=1,107 | 0.066 [0.101] N=987 | 0.088 [0.111] N=860 | 0.039 [0.129] N=716 |
| Foreign language | 3.670 | 0.047 [0.101] N=1,343 | 0.129 [0.106] N=1,236 | 0.118 [0.117] N=1,113 | 0.134 [0.130] N=974 | 0.137 [0.147] N=812 |
| Culture arts and sports | 3.996 | 0.043 [0.064] N=1,508 | 0.059 [0.067] N=1,390 | 0.022 [0.069] N=1,256 | 0.006 [0.079] N=1,095 | -0.046 [0.088] N=919 |
| Vocational and technological | 3.837 | 0.040 [0.137] N=563 | 0.063 [0.145] N=512 | 0.083 [0.159] N=466 | 0.087 [0.166] N=411 | 0.011 [0.185] N=338 |

*Specification 6 Monthly fixed effects female and any aid (0/1) 12 month donut hole*

| Outcome | 8 years | 7 years | 6 years | 5 years | 4 years |
|---|---|---|---|---|---|
| Turkish | -0.101 [0.086] N=1,520 | -0.053 [0.093] N=1,384 | 0.000 [0.104] N=1,217 | 0.009 [0.117] N=1,024 | -0.150 [0.135] N=810 |
| Social studies and life sciences | 0.032 [0.081] N=1,408 | 0.074 [0.087] N=1,280 | 0.099 [0.099] N=1,124 | 0.109 [0.111] N=950 | -0.030 [0.119] N=752 |
| Mathematics | 0.048 [0.097] N=1,492 | 0.147 [0.106] N=1,358 | 0.166 [0.119] N=1,193 | **0.231* [0.138] N=1,005** | 0.154 [0.171] N=800 |
| Science | -0.128 [0.111] N=1,026 | -0.086 [0.121] N=926 | -0.057 [0.134] N=806 | -0.087 [0.157] N=679 | -0.219 [0.184] N=535 |
| Foreign language | -0.047 [0.123] N=1,143 | 0.083 [0.135] N=1,036 | 0.025 [0.152] N=913 | 0.033 [0.174] N=774 | -0.008 [0.196] N=612 |
| Culture arts and sports | 0.056 [0.099] N=1,274 | 0.084 [0.110] N=1,156 | 0.024 [0.121] N=1,022 | 0.008 [0.154] N=861 | -0.104 [0.199] N=685 |
| Vocational and technological | 0.110 [0.181] N=487 | 0.174 [0.195] N=436 | 0.211 [0.232] N=390 | 0.232 [0.258] N=335 | 0.134 [0.341] N=262 |

## Mother RDD secondary analysis

*Specification 1 Monthly child-age fixed effects  No donut hole*

| Outcome | Mean | 8 years | 7 years | 6 years | 5 years | 4 years |
|---|---|---|---|---|---|---|
| Turkish | 3.931 | 0.090 [0.060]<br>N=2,350 | 0.090 [0.064]<br>N=2,146 | 0.075 [0.069]<br>N=1,902 | 0.062 [0.076]<br>N=1,637 | 0.052 [0.083]<br>N=1,321 |
| Social studies and life sciences | 3.946 | 0.089 [0.057]<br>N=2,202 | 0.080 [0.060]<br>N=2,012 | 0.068 [0.067]<br>N=1,787 | 0.045 [0.073]<br>N=1,534 | 0.053 [0.081]<br>N=1,239 |
| Mathematics | 3.883 | 0.018 [0.056]<br>N=2,313 | 0.023 [0.057]<br>N=2,113 | 0.013 [0.062]<br>N=1,875 | -0.022 [0.067]<br>N=1,615 | 0.042 [0.075]<br>N=1,304 |
| Science | 3.892 | 0.020 [0.065]<br>N=1,631 | 0.021 [0.067]<br>N=1,498 | 0.022 [0.074]<br>N=1,329 | 0.009 [0.080]<br>N=1,147 | 0.036 [0.088]<br>N=934 |
| Foreign language | 3.692 | 0.065 [0.070]<br>N=1,805 | 0.042 [0.074]<br>N=1,658 | 0.020 [0.080]<br>N=1,474 | 0.031 [0.086]<br>N=1,268 | 0.005 [0.085]<br>N=1,029 |
| Culture arts and sports | 4.009 | 0.005 [0.053]<br>N=1,961 | 0.000 [0.055]<br>N=1,794 | 0.020 [0.058]<br>N=1,597 | -0.004 [0.061]<br>N=1,368 | 0.006 [0.071]<br>N=1,116 |
| Vocational and technological | 3.855 | 0.074 [0.118]<br>N=757 | 0.176 [0.121]<br>N=686 | **0.230* [0.131]<br>N=601** | **0.277* [0.146]<br>N=517** | 0.276 [0.167]<br>N=427 |

*Specification 1 Monthly child-age fixed effects  12 month donut hole*

| Outcome | 8 years | 7 years | 6 years | 5 years | 4 years |
|---|---|---|---|---|---|
| Turkish | **0.135* [0.075]<br>N=2,000** | **0.140* [0.082]<br>N=1,796** | 0.124 [0.094]<br>N=1,552 | 0.107 [0.103]<br>N=1,287 | 0.122 [0.122]<br>N=971 |
| Social studies and life sciences | **0.130* [0.074]<br>N=1,874** | 0.114 [0.080]<br>N=1,684 | 0.095 [0.096]<br>N=1,459 | 0.054 [0.107]<br>N=1,206 | 0.091 [0.128]<br>N=911 |
| Mathematics | 0.000 [0.078]<br>N=1,966 | -0.005 [0.081]<br>N=1,766 | -0.032 [0.093]<br>N=1,528 | -0.127 [0.107]<br>N=1,268 | 0.011 [0.131]<br>N=957 |
| Science | 0.000 [0.093]<br>N=1,383 | -0.016 [0.098]<br>N=1,250 | -0.019 [0.117]<br>N=1,081 | -0.079 [0.131]<br>N=899 | -0.035 [0.162]<br>N=686 |
| Foreign language | 0.050 [0.099]<br>N=1,522 | 0.003 [0.109]<br>N=1,375 | -0.049 [0.128]<br>N=1,191 | -0.068 [0.152]<br>N=985 | -0.171 [0.176]<br>N=746 |
| Culture arts and sports | -0.041 [0.073]<br>N=1,658 | -0.069 [0.079]<br>N=1,491 | -0.042 [0.090]<br>N=1,294 | -0.123 [0.096]<br>N=1,065 | -0.141 [0.134]<br>N=813 |
| Vocational and technological | -0.070 [0.146]<br>N=650 | 0.067 [0.152]<br>N=579 | 0.108 [0.168]<br>N=494 | 0.168 [0.197]<br>N=410 | 0.228 [0.239]<br>N=320 |

*Specification 2 Monthly fixed effects and female  No donut hole*

| Outcome | Mean | 8 years | 7 years | 6 years | 5 years | 4 years |
|---|---|---|---|---|---|---|
| Turkish | 3.931 | 0.084 [0.059] N=2,350 | 0.086 [0.063] N=2,146 | 0.073 [0.068] N=1,902 | 0.064 [0.075] N=1,637 | 0.057 [0.082] N=1,321 |
| Social studies and life sciences | 3.946 | 0.087 [0.056] N=2,202 | 0.079 [0.060] N=2,012 | 0.069 [0.066] N=1,787 | 0.048 [0.073] N=1,534 | 0.058 [0.081] N=1,239 |
| Mathematics | 3.883 | 0.020 [0.056] N=2,313 | 0.024 [0.057] N=2,113 | 0.013 [0.062] N=1,875 | -0.023 [0.067] N=1,615 | 0.041 [0.075] N=1,304 |
| Science | 3.892 | 0.019 [0.065] N=1,631 | 0.021 [0.067] N=1,498 | 0.023 [0.074] N=1,329 | 0.013 [0.081] N=1,147 | 0.043 [0.089] N=934 |
| Foreign language | 3.692 | 0.064 [0.070] N=1,805 | 0.042 [0.074] N=1,658 | 0.023 [0.080] N=1,474 | 0.042 [0.086] N=1,268 | 0.017 [0.085] N=1,029 |
| Culture arts and sports | 4.009 | 0.002 [0.054] N=1,961 | -0.002 [0.055] N=1,794 | 0.019 [0.059] N=1,597 | -0.004 [0.061] N=1,368 | 0.007 [0.071] N=1,116 |
| Vocational and technological | 3.855 | 0.065 [0.119] N=757 | 0.168 [0.122] N=686 | **0.226* [0.132] N=601** | **0.270* [0.147] N=517** | 0.268 [0.168] N=427 |

*Specification 2 Monthly fixed effects and female  12 month donut hole*

| Outcome | 8 years | 7 years | 6 years | 5 years | 4 years |
|---|---|---|---|---|---|
| Turkish | 0.116 [0.075] N=2,000 | 0.121 [0.082] N=1,796 | 0.106 [0.094] N=1,552 | 0.083 [0.104] N=1,287 | 0.098 [0.123] N=971 |
| Social studies and life sciences | 0.116 [0.074] N=1,874 | 0.102 [0.080] N=1,684 | 0.085 [0.096] N=1,459 | 0.043 [0.107] N=1,206 | 0.076 [0.128] N=911 |
| Mathematics | 0.008 [0.078] N=1,966 | 0.003 [0.082] N=1,766 | -0.028 [0.093] N=1,528 | -0.122 [0.108] N=1,268 | 0.016 [0.131] N=957 |
| Science | -0.006 [0.093] N=1,383 | -0.021 [0.098] N=1,250 | -0.026 [0.116] N=1,081 | -0.086 [0.131] N=899 | -0.042 [0.163] N=686 |
| Foreign language | 0.039 [0.098] N=1,522 | -0.009 [0.108] N=1,375 | -0.059 [0.126] N=1,191 | -0.074 [0.150] N=985 | -0.178 [0.175] N=746 |
| Culture arts and sports | -0.050 [0.074] N=1,658 | -0.077 [0.080] N=1,491 | -0.049 [0.090] N=1,294 | -0.134 [0.096] N=1,065 | -0.156 [0.133] N=813 |
| Vocational and technological | -0.088 [0.148] N=650 | 0.051 [0.152] N=579 | 0.098 [0.167] N=494 | 0.143 [0.195] N=410 | 0.204 [0.239] N=320 |

*Specification 3 Monthly fixed effects female and household income groups No donut hole*

| Outcome | Mean | 8 years | 7 years | 6 years | 5 years | 4 years |
|---|---|---|---|---|---|---|
| Turkish | 3.931 | 0.084 [0.057] N=2,350 | 0.084 [0.062] N=2,146 | 0.070 [0.065] N=1,902 | 0.059 [0.071] N=1,637 | 0.052 [0.079] N=1,321 |
| Social studies and life sciences | 3.946 | **0.091* [0.053] N=2,202** | 0.083 [0.056] N=2,012 | 0.072 [0.061] N=1,787 | 0.052 [0.066] N=1,534 | 0.060 [0.075] N=1,239 |
| Mathematics | 3.883 | 0.022 [0.055] N=2,313 | 0.026 [0.057] N=2,113 | 0.012 [0.060] N=1,875 | -0.025 [0.066] N=1,615 | 0.038 [0.075] N=1,304 |
| Science | 3.892 | 0.022 [0.063] N=1,631 | 0.025 [0.065] N=1,498 | 0.029 [0.072] N=1,329 | 0.019 [0.076] N=1,147 | 0.048 [0.086] N=934 |
| Foreign language | 3.692 | 0.062 [0.069] N=1,805 | 0.045 [0.074] N=1,658 | 0.026 [0.078] N=1,474 | 0.045 [0.084] N=1,268 | 0.021 [0.085] N=1,029 |
| Culture arts and sports | 4.009 | 0.008 [0.051] N=1,961 | 0.006 [0.053] N=1,794 | 0.025 [0.056] N=1,597 | 0.004 [0.058] N=1,368 | 0.014 [0.067] N=1,116 |
| Vocational and technological | 3.855 | 0.067 [0.114] N=757 | 0.165 [0.117] N=686 | **0.245* [0.127] N=601** | **0.302** [0.140] N=517** | **0.325** [0.155] N=427** |

*Specification 3 Monthly fixed effects female and household income groups 12 month donut hole*

| Outcome | 8 years | 7 years | 6 years | 5 years | 4 years |
|---|---|---|---|---|---|
| Turkish | 0.118 [0.074] N=2,000 | 0.123 [0.082] N=1,796 | 0.109 [0.091] N=1,552 | 0.087 [0.100] N=1,287 | 0.105 [0.117] N=971 |
| Social studies and life sciences | **0.123* [0.071] N=1,874** | 0.111 [0.078] N=1,684 | 0.094 [0.091] N=1,459 | 0.058 [0.100] N=1,206 | 0.094 [0.122] N=911 |
| Mathematics | 0.012 [0.076] N=1,966 | 0.009 [0.081] N=1,766 | -0.024 [0.089] N=1,528 | -0.116 [0.105] N=1,268 | 0.024 [0.130] N=957 |
| Science | -0.002 [0.092] N=1,383 | -0.010 [0.098] N=1,250 | -0.007 [0.116] N=1,081 | -0.058 [0.130] N=899 | -0.004 [0.167] N=686 |
| Foreign language | 0.042 [0.095] N=1,522 | 0.006 [0.106] N=1,375 | -0.039 [0.120] N=1,191 | -0.051 [0.147] N=985 | -0.149 [0.171] N=746 |
| Culture arts and sports | -0.048 [0.075] N=1,658 | -0.072 [0.081] N=1,491 | -0.047 [0.089] N=1,294 | -0.132 [0.097] N=1,065 | -0.149 [0.135] N=813 |
| Vocational and technological | -0.112 [0.142] N=650 | 0.020 [0.145] N=579 | 0.105 [0.160] N=494 | 0.148 [0.190] N=410 | 0.303 [0.216] N=320 |

*Specification 4 Monthly fixed effects female and per-person income groups  No donut hole*

| Outcome | Mean | 8 years | 7 years | 6 years | 5 years | 4 years |
|---|---|---|---|---|---|---|
| Turkish | 3.931 | 0.076 [0.057]<br>N=2,350 | 0.073 [0.061]<br>N=2,146 | 0.057 [0.065]<br>N=1,902 | 0.047 [0.070]<br>N=1,637 | 0.033 [0.078]<br>N=1,321 |
| Social studies and life sciences | 3.946 | 0.084 [0.052]<br>N=2,202 | 0.072 [0.056]<br>N=2,012 | 0.060 [0.060]<br>N=1,787 | 0.040 [0.065]<br>N=1,534 | 0.043 [0.073]<br>N=1,239 |
| Mathematics | 3.883 | 0.013 [0.056]<br>N=2,313 | 0.012 [0.058]<br>N=2,113 | -0.002 [0.061]<br>N=1,875 | -0.036 [0.066]<br>N=1,615 | 0.023 [0.076]<br>N=1,304 |
| Science | 3.892 | 0.010 [0.063]<br>N=1,631 | 0.007 [0.066]<br>N=1,498 | 0.008 [0.072]<br>N=1,329 | 0.002 [0.076]<br>N=1,147 | 0.017 [0.085]<br>N=934 |
| Foreign language | 3.692 | 0.049 [0.069]<br>N=1,805 | 0.028 [0.074]<br>N=1,658 | 0.011 [0.078]<br>N=1,474 | 0.035 [0.086]<br>N=1,268 | -0.001 [0.085]<br>N=1,029 |
| Culture arts and sports | 4.009 | -0.001 [0.051]<br>N=1,961 | -0.008 [0.052]<br>N=1,794 | 0.011 [0.055]<br>N=1,597 | -0.001 [0.056]<br>N=1,368 | -0.001 [0.065]<br>N=1,116 |
| Vocational and technological | 3.855 | 0.045 [0.109]<br>N=757 | 0.142 [0.112]<br>N=686 | **0.229* [0.119]<br>N=601** | **0.284** [0.134]<br>N=517** | **0.287* [0.154]<br>N=427** |

*Specification 4 Monthly fixed effects female and per-person income groups  12 month donut hole*

| Outcome | 8 years | 7 years | 6 years | 5 years | 4 years |
|---|---|---|---|---|---|
| Turkish | **0.121* [0.073]<br>N=2,000** | 0.124 [0.081]<br>N=1,796 | 0.107 [0.092]<br>N=1,552 | 0.095 [0.100]<br>N=1,287 | 0.096 [0.119]<br>N=971 |
| Social studies and life sciences | **0.123* [0.071]<br>N=1,874** | 0.108 [0.077]<br>N=1,684 | 0.090 [0.090]<br>N=1,459 | 0.060 [0.099]<br>N=1,206 | 0.087 [0.123]<br>N=911 |
| Mathematics | 0.012 [0.078]<br>N=1,966 | 0.003 [0.082]<br>N=1,766 | -0.030 [0.091]<br>N=1,528 | -0.117 [0.106]<br>N=1,268 | 0.016 [0.136]<br>N=957 |
| Science | -0.005 [0.091]<br>N=1,383 | -0.022 [0.098]<br>N=1,250 | -0.029 [0.116]<br>N=1,081 | -0.066 [0.129]<br>N=899 | -0.033 [0.172]<br>N=686 |
| Foreign language | 0.033 [0.095]<br>N=1,522 | -0.008 [0.105]<br>N=1,375 | -0.053 [0.120]<br>N=1,191 | -0.051 [0.147]<br>N=985 | -0.161 [0.173]<br>N=746 |
| Culture arts and sports | -0.047 [0.073]<br>N=1,658 | -0.078 [0.078]<br>N=1,491 | -0.054 [0.088]<br>N=1,294 | -0.119 [0.096]<br>N=1,065 | -0.154 [0.135]<br>N=813 |
| Vocational and technological | -0.125 [0.136]<br>N=650 | 0.005 [0.140]<br>N=579 | 0.107 [0.149]<br>N=494 | 0.175 [0.174]<br>N=410 | 0.272 [0.211]<br>N=320 |

*Specification 5 Monthly fixed effects female and number of aid types (0-6) No donut hole*

| Outcome | Mean | 8 years | 7 years | 6 years | 5 years | 4 years |
|---|---|---|---|---|---|---|
| Turkish | 3.922 | **0.101* [0.061] N=2,215** | **0.111* [0.065] N=2,020** | 0.100 [0.069] N=1,787 | 0.087 [0.075] N=1,536 | 0.067 [0.083] N=1,244 |
| Social studies and life sciences | 3.936 | **0.109* [0.058] N=2,080** | **0.108* [0.061] N=1,899** | 0.104 [0.068] N=1,685 | 0.087 [0.075] N=1,444 | 0.079 [0.084] N=1,171 |
| Mathematics | 3.872 | 0.028 [0.058] N=2,183 | 0.038 [0.058] N=1,992 | 0.044 [0.063] N=1,765 | 0.012 [0.069] N=1,518 | 0.070 [0.080] N=1,231 |
| Science | 3.880 | 0.026 [0.070] N=1,535 | 0.019 [0.072] N=1,410 | 0.033 [0.079] N=1,250 | 0.023 [0.086] N=1,077 | 0.026 [0.097] N=882 |
| Foreign language | 3.673 | 0.074 [0.072] N=1,707 | 0.049 [0.075] N=1,568 | 0.039 [0.080] N=1,391 | 0.063 [0.085] N=1,196 | 0.022 [0.086] N=976 |
| Culture arts and sports | 4.008 | 0.030 [0.055] N=1,880 | 0.021 [0.056] N=1,720 | 0.035 [0.059] N=1,527 | 0.009 [0.061] N=1,305 | 0.020 [0.073] N=1,069 |
| Vocational and technological | 3.856 | 0.113 [0.116] N=728 | **0.199* [0.120] N=661** | **0.263** [0.129] N=577** | **0.306** [0.141] N=498** | **0.349** [0.160] N=413** |

*Specification 5 Monthly fixed effects female and number of aid types (0-6) 12 month donut hole*

| Outcome | 8 years | 7 years | 6 years | 5 years | 4 years |
|---|---|---|---|---|---|
| Turkish | **0.135* [0.077] N=1,882** | **0.155* [0.083] N=1,687** | 0.146 [0.093] N=1,454 | 0.125 [0.103] N=1,203 | 0.103 [0.129] N=911 |
| Social studies and life sciences | **0.137* [0.076] N=1,766** | 0.133 [0.081] N=1,585 | 0.129 [0.097] N=1,371 | 0.104 [0.107] N=1,130 | 0.082 [0.131] N=857 |
| Mathematics | 0.007 [0.079] N=1,851 | 0.008 [0.081] N=1,660 | 0.007 [0.094] N=1,433 | -0.069 [0.111] N=1,186 | 0.056 [0.135] N=899 |
| Science | -0.001 [0.099] N=1,298 | -0.033 [0.101] N=1,173 | -0.023 [0.120] N=1,013 | -0.070 [0.135] N=840 | -0.123 [0.163] N=645 |
| Foreign language | 0.066 [0.104] N=1,435 | 0.016 [0.115] N=1,296 | -0.019 [0.132] N=1,119 | 0.004 [0.150] N=924 | -0.129 [0.179] N=704 |
| Culture arts and sports | 0.002 [0.079] N=1,587 | -0.030 [0.084] N=1,427 | -0.008 [0.093] N=1,234 | -0.089 [0.099] N=1,012 | -0.109 [0.139] N=776 |
| Vocational and technological | -0.021 [0.152] N=623 | 0.092 [0.156] N=556 | 0.136 [0.173] N=472 | 0.184 [0.206] N=393 | 0.307 [0.250] N=308 |

*Specification 6 Monthly fixed effects female and any aid (0/1) No donut hole*

| Outcome | Mean | 8 years | 7 years | 6 years | 5 years | 4 years |
|---|---|---|---|---|---|---|
| Turkish | 3.922 | 0.097 [0.061] N=2,215 | 0.106 [0.065] N=2,020 | 0.093 [0.069] N=1,787 | 0.077 [0.076] N=1,536 | 0.060 [0.084] N=1,244 |
| Social studies and life sciences | 3.936 | **0.106* [0.058] N=2,080** | **0.104* [0.061] N=1,899** | 0.098 [0.068] N=1,685 | 0.078 [0.075] N=1,444 | 0.070 [0.083] N=1,171 |
| Mathematics | 3.872 | 0.023 [0.058] N=2,183 | 0.033 [0.059] N=1,992 | 0.036 [0.064] N=1,765 | 0.001 [0.069] N=1,518 | 0.063 [0.078] N=1,231 |
| Science | 3.880 | 0.022 [0.071] N=1,535 | 0.015 [0.072] N=1,410 | 0.026 [0.079] N=1,250 | 0.014 [0.086] N=1,077 | 0.017 [0.095] N=882 |
| Foreign language | 3.673 | 0.073 [0.071] N=1,707 | 0.048 [0.075] N=1,568 | 0.038 [0.080] N=1,391 | 0.062 [0.084] N=1,196 | 0.024 [0.085] N=976 |
| Culture arts and sports | 4.008 | 0.026 [0.055] N=1,880 | 0.016 [0.056] N=1,720 | 0.030 [0.059] N=1,527 | -0.001 [0.062] N=1,305 | 0.005 [0.072] N=1,069 |
| Vocational and technological | 3.856 | 0.117 [0.119] N=728 | 0.203 [0.123] N=661 | **0.254* [0.132] N=577** | **0.291** [0.147] N=498** | **0.308* [0.168] N=413** |

*Specification 6 Monthly fixed effects female and any aid (0/1) 12 month donut hole*

| Outcome | 8 years | 7 years | 6 years | 5 years | 4 years |
|---|---|---|---|---|---|
| Turkish | **0.129* [0.077] N=1,882** | **0.148* [0.084] N=1,687** | 0.137 [0.095] N=1,454 | 0.105 [0.105] N=1,203 | 0.090 [0.129] N=911 |
| Social studies and life sciences | **0.135* [0.076] N=1,766** | 0.131 [0.081] N=1,585 | 0.124 [0.097] N=1,371 | 0.090 [0.108] N=1,130 | 0.076 [0.128] N=857 |
| Mathematics | -0.002 [0.079] N=1,851 | -0.001 [0.081] N=1,660 | -0.004 [0.094] N=1,433 | -0.093 [0.110] N=1,186 | 0.043 [0.133] N=899 |
| Science | -0.003 [0.100] N=1,298 | -0.034 [0.102] N=1,173 | -0.025 [0.120] N=1,013 | -0.073 [0.134] N=840 | -0.109 [0.162] N=645 |
| Foreign language | 0.064 [0.105] N=1,435 | 0.014 [0.115] N=1,296 | -0.020 [0.132] N=1,119 | -0.002 [0.151] N=924 | -0.118 [0.182] N=704 |
| Culture arts and sports | 0.001 [0.079] N=1,587 | -0.031 [0.084] N=1,427 | -0.010 [0.095] N=1,234 | -0.098 [0.101] N=1,012 | -0.116 [0.143] N=776 |
| Vocational and technological | -0.009 [0.151] N=623 | 0.106 [0.154] N=556 | 0.147 [0.171] N=472 | 0.199 [0.201] N=393 | 0.306 [0.248] N=308 |

*Notes:* Cells report the reform eligibility coefficient, parental birth-month-clustered standard error in brackets, and N. All models use exact child-month fixed effects calculated on 16 December 2022, split linear trends at January 1987, and FAKTOR_FERT child survey weights. Specification 2 adds female; specifications 3 and 4 separately add five household-income and five per-person-income groups; specifications 5 and 6 separately add categorical aid count and any aid. Aid specifications require valid yes/no answers to all six household aid questions. Income and aid controls are separate alternatives. Columns use 8- to 4-year bandwidths. Donuts omit running months -12 through +11. No child-age interval restriction is imposed; complete birth dates are needed for sibling selection and monthly controls. P is the unweighted positive-case count in the no-donut 8-year sample. NA denotes a failed or zero-standard-error fit. These are caregiver assessments. School-level and subject-response selection can affect interpretation. *** p<0.01, ** p<0.05, * p<0.10.